

# What helps people learn

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*Citations from a reference file*

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# Less on the slide

Working memory holds only a few things at once, so a slide that makes people search wastes it (Sweller, 1988).

Words spoken alongside a picture beat the same words printed on it (Mayer, 2009, Chapter 11).

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Sweller, J. (1988). Cognitive Load During Problem Solving: Effects on Learning. *Cognitive Science*, 12(2), 257–285. [https://doi.org/10.1207/s15516709cog1202\\_4](https://doi.org/10.1207/s15516709cog1202_4)

Mayer, R. E. (2009). *Multimedia Learning* (2nd ed.). Cambridge University Press.

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# Ask, don't tell

As Roediger & Karpicke (2006) showed, recalling something fixes it better than reading it again.

Lectures with active learning cut failure rates by about a third (Freeman et al., 2014; see also Roediger & Karpicke, 2006).

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Roediger, H. L., & Karpicke, J. D. (2006). Test-Enhanced Learning: Taking Memory Tests Improves Long-Term Retention. *Psychological Science*, 17(3), 249–255. <https://doi.org/10.1111/j.1467-9280.2006.01693.x>

Freeman, S., Eddy, S. L., McDonough, M., Smith, M. K., Okoroafor, N., Jordt, H., & Wenderoth, M. P. (2014). Active learning increases student performance in science, engineering, and mathematics. *Proceedings of the National Academy of Sciences*, 111(23), 8410–8415. <https://doi.org/10.1073/pnas.1319030111>

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# References

- Freeman, S., Eddy, S. L., McDonough, M., Smith, M. K., Okoroafor, N., Jordt, H., & Wenderoth, M. P. (2014). Active learning increases student performance in science, engineering, and mathematics. *Proceedings of the National Academy of Sciences*, 111(23), 8410–8415. <https://doi.org/10.1073/pnas.1319030111>
- Mayer, R. E. (2009). *Multimedia Learning* (2nd ed.). Cambridge University Press.
- Roediger, H. L., & Karpicke, J. D. (2006). Test-Enhanced Learning: Taking Memory Tests Improves Long-Term Retention. *Psychological Science*, 17(3), 249–255. <https://doi.org/10.1111/j.1467-9280.2006.01693.x>
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